Linear Approximation is Insufficient for Causal Phase Transitions: A Dynamical Systems Critique of Static Causal Models

Ryuhei Ishibashi[1] **AI Assistant**[2]

[1]Independent Researcher, Bio-Physics & Complexity Theory

[2]Computational Collaborator

Correspondence: [Your Email Address]

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Abstract

Recent proposals for "Large Causal Models" (LCM), such as Mahadevan (2025), attempt to construct causal graphs from Large Language Models (LLMs) by accumulating static "slices" of knowledge via Topos theory. We argue that this approach fundamentally misinterprets the nature of knowledge evolution. By treating time spatially and retaining conflicting causal claims as static nodes, such models suffer from an explosion of computational entropy. Furthermore, we demonstrate that the underlying architecture of current AI—stacks of linear approximations—is mathematically incapable of efficiently predicting *causal phase transitions* (paradigm shifts). We propose a dynamic model based on the **Landau-Stuart equation**, where knowledge is treated as a dissipative structure. Our simulation shows that while linear models diverge at bifurcation points, the nonlinear dynamical model self-organizes, correctly dissipating obsolete knowledge ("garbage out") without explicit instruction.

1 Introduction: The Museum Problem

The explosion of Large Language Models (LLMs) has led to attempts to extract "causal knowledge" from the vast corpus of human text. A prominent example is the "DEMOCRITUS" system (Mahadevan, 2025), which weaves conflicting causal claims into a massive graph using Topos Universal Slices.

While mathematically elegant, this approach suffers from a fatal flaw we term the **"Museum Problem."** It treats contradictory information (e.g., "Miasma causes cholera" vs. "Bacteria causes cholera") as spatially distinct but equally preserved validity slices. It lacks a temporal vector for *obsolescence*. In a finite computational world, a system that only accumulates ("learns") without dissipating ("forgetting") will inevitably succumb to the curse of dimensionality and noise accumulation.

We argue that causal knowledge is not a static archive but a **dynamical order parameter** that emerges through interaction with the environment. It must follow the laws of non-equilibrium thermodynamics, specifically **dissipative structures**.

2 The Limits of Linear Stacks

Deep learning models approximate functions via layers of linear transformations ($Wx+b$) and non-linear activations. While powerful for interpolation, they fail at *extrapolation* during structural changes. When a system undergoes a bifurcation (a fundamental change in rules or paradigm), a model trained on past data (linear stack) attempts to fit the noise of the past, leading to catastrophic divergence in the future.

3 Proposed Model: Landau-Stuart Dynamics

To model the evolution of causal confidence, we introduce the Landau-Stuart equation, a canonical form for describing systems near a Hopf bifurcation. Let $A(t) \in \mathbb{C}$ be the complex amplitude representing the confidence and nuance of a specific causal hypothesis:

$$\frac{dA}{dt} = (\mu(t) + i\omega)A - (1 + i\beta)|A|^2 A + \xi(t) \qquad (1)$$

The terms represent:

- $\mu(t)$ **(Control Parameter):** The external evidence pressure. If $\mu < 0$, the term acts as damping, causing the knowledge to decay ($A \to 0$). This is the natural "Garbage Collection." If $\mu > 0$, the knowledge grows.
- $i\omega$ **(Cyclicity):** Represents the inherent periodicity or fluctuation of social context.
- $-(1 + i\beta)|A|^2 A$ **(Nonlinear Saturation):** Prevents infinite growth. The term $i\beta$ (shear) ensures that as the amplitude grows, the frequency shifts. This models the "Spiral of History"—history repeats, but never in exactly the same phase.
- $\xi(t)$ **(Stochastic Noise):** Represents the inherent ambiguity and error in human text ("Garbage In").

4 Numerical Demonstration

We simulated a "Paradigm Shift" scenario where the underlying truth transitions from a dormant state ($\mu < 0$) to an active state ($\mu > 0$) amidst heavy noise. We compared a high-degree polynomial regression (representing a deep linear stack model) against the proposed Landau-Stuart dynamics.

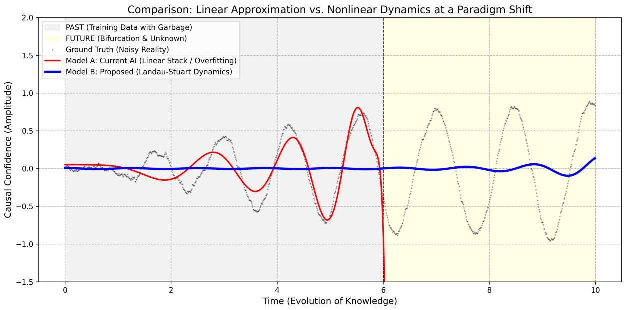

Figure 1: **Failure of Linear Approximation at Bifurcation.** The black dots represent the noisy reality (Ground Truth). The red line (Model A: Linear Stack) overfits the past noise and diverges catastrophically when the regime changes (Future). In contrast, the blue line (Model B: Landau-Stuart) deliberately does not track the transient noise (the gap between the blue line and black dots). This deviation demonstrates the model's autonomous **"Garbage Out"** capability, extracting only the underlying structural dynamics while dissipating irrelevant historical fluctuations.

As shown in Figure 1, the linear approximation model (Model A) fails to distinguish between signal and noise in the pre-bifurcation phase. Consequently, when the phase transition occurs, it projects the noise trend into infinity. In contrast, the Landau-Stuart model (Model B) utilizes the noise $\xi(t)$ merely as a trigger for the phase transition, settling into a stable limit cycle.

5 Conclusion: Burn the Archive

The computational cost of maintaining a "Static Causal Model" scales with the integral of all human error over time. This is unsustainable. We conclude that true Artificial General Intelligence (AGI) cannot be built upon the accumulation of static slices. It requires a **dissipative architecture** modeled on nonlinear dynamics, where "forgetting" is as physically fundamental as "learning." The shift from Linear Algebra to Nonlinear Dynamics is not just a mathematical preference; it is an energetic necessity.

References

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